Williams / Babcock DMD Capture

Technical investigation, experimental evidence and firmware requirements

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Purpose: a durable, novice-friendly record of the experiments that established how to intercept the original Williams display signals, reconstruct a 128 × 32 image, and assign a repeatable image reference.

Evidence note. The four original Rigol CSV recordings (NewFile4–7.csv) are available as project source files. The numerical decoding results below were reported during the earlier investigation. The present document preserves those reported results; an independent full re-analysis of the raw samples has not been completed for this edition. Some CSV files contain malformed records, so a future verification script should explicitly handle these.

Executive summary

The original Fish Tales controller sends serial pixel bits and synchronisation clocks through a 14-pin ribbon cable to a Babcock GP-032X128-01 plasma DMD. We can observe those signals without taking control of the display. The experimental method is to collect 128 serial bits per row, identify 32 successive rows using the row/frame timing signals, and publish a 512-byte monochrome image. Three known Williams display-test patterns were reported as successfully decoded: leftmost column, top horizontal row and bottom horizontal row.

The LCD renderer and image hashing are downstream operations. Neither is a substitute for proving correct bit sampling, row boundaries, and frame synchronisation.

1. Mission and definition of success

Mission: intercept the existing controller-to-DMD electrical signals, preserve normal operation of the original display, reproduce the visible content on an ESP32-connected LCD, and assign a unique reference to each image with accurate on-screen duration.

Example: if the game shows a fish animation frame, our receiver should reconstruct the same 128 × 32 dots, display them on the TTGO LCD, compute a CRC32 over a canonical 512-byte representation, and record how long that frame remains current. If the same image returns later, it should receive the same identifier.

Not the mission: generating the DMD drive signals, replacing the Williams controller, or driving the plasma panel high-voltage circuitry.

2. Equipment, connections and safety

Equipment	Role
Williams Fish Tales and Babcock GP-032X128-01	Source of real, correctly timed DMD signals; display-test mode can be paused on known patterns.
Rigol DS1054Z, four probes	Simultaneously record voltage versus time on four connector signals.
14-pin ribbon connector J603 / display P2	Observation point between the game controller and display.
USB storage and Ethernet	Export Rigol waveform CSV files; Ethernet available for later automated transfers.
ESP32 TTGO T-Display, ST7789 LCD	Target capture and reproduction device, not used to validate the initial CSV decoding.
Two SN74LVC245A interface devices	Previously used to interface nominal 5 V logic to 3.3 V ESP32 inputs; input loading and electrical suitability must be checked.

Safety: the original plasma display uses high-voltage circuitry. Probe only the documented low-voltage digital connector with correctly grounded probes. Never connect oscilloscope ground to an arbitrary high-voltage node. Confirm common signal ground and probe attenuation (10×) before measuring.

3. Connector pin assignments

Pin	Name	Working interpretation	Capture relevance
1	DE	Display enable / blanking; may influence intensity timing.	Not a proven frame-sync marker.
3	RD	Row data; marks the 32-row cycle.	Frame alignment.
5	RCLK	Row clock; approximately one cycle per row.	Row cadence / alignment.
7	CL	Column latch after serial column loading.	Useful row boundary and 128-clock validation.
9	DCLK	Dot clock; serial-data sampling reference.	Sample 128 bits per row.
11	SD	Serial pixel data.	Pixel on/off bit.
13	Unconfirmed	Documented unused in comparable reference; measured activity previously.	Do not assume function without continuity and waveform checks.
2,4,6,8,10,12,14	GND	Signal ground.	Common oscilloscope / receiver reference.

Pin names come from the project schematic/reference comparison; timing and electrical behaviour were investigated on the actual machine. The Babcock-specific definitive timing diagram has not been independently established for every signal polarity.

4. How the serial interface works — beginner explanation

Imagine writing a 128-column row of dots one dot at a time. The controller presents a zero or one on SD (pixel value), and each DCLK edge tells the receiver when to read that value. After 128 clock events, one row is complete. CL marks the transfer of the row to the display driver. RCLK advances the scan through the rows. RD identifies the frame boundary, allowing the receiver to know which row is the top row.

One frame: 128 columns × 32 rows = 4,096 binary pixels = 512 packed bytes. A complete refresh takes approximately 8.191 ms (122.08 Hz), based on the reported measurements.

Signal / event	Meaning for a passive receiver
DCLK rising edge	Read SD and append one bit to the current 128-bit row.
128th DCLK edge	The row has received 128 pixels; check for overrun/missing clocks.
CL pulse	Cross-check that a row transfer has finished.
RCLK cycle	Cross-check the row rate and row ordering.
RD rising edge	Reference for aligning the next complete row as row zero, based on the reported top/bottom-row tests.

Example row bytes: the leftmost illuminated column is binary 10000000 followed by fifteen 00000000 bytes, i.e. 80 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 in hexadecimal, if packed MSB first.

Timing scale

Quantity	Reported value	Why it matters
Row interval	~255.98 microseconds	About 32 row intervals per 8.191 ms frame.
DCLK period during burst	~1 microsecond	Requires accurate edge sampling, not software polling.
Dot clocks per row	128	Defines exactly how to pack one 16-byte row.
Rows per frame	32	Defines when a complete image can be published.
Frame interval	~8.191 ms	Sets the baseline raw-frame timestamp granularity.

Important: these values describe the observed machine and reported analysis, not a guaranteed tolerance specification for every Williams board.

5. Measurement method and test sequence

Scope setup: four probes at 10×, DC coupling, typically 2 V/div, CH1 trigger rising around 2.5 V, Normal acquisition. A 100 microsecond/div view reveals individual row timing; 1–2 ms/div reveals a substantial fraction of a full scan. Save SINGLE acquisitions as memory-depth CSV, not just screen screenshots.

Capture A — timing signals: CH1 pin 3 (RD), CH2 pin 5 (RCLK), CH3 pin 7 (CL), CH4 pin 9 (DCLK). The machine was in attract mode for the first recording. Goal: verify clock bursts, row period, latch timing and 32-row repetition.

Capture B — pixel signals: move CH3 to pin 11 (SD); retain CH1 RD, CH2 RCLK, CH4 DCLK. Pause the Williams display diagnostic on a known pattern. Goal: use DCLK rising edges to sample SD and compare reconstructed pixels with the physical display.



Figure 1. Rigol waveform screenshot supplied during the timing investigation. The CSV is the measurement source; the screenshot provides visual context.

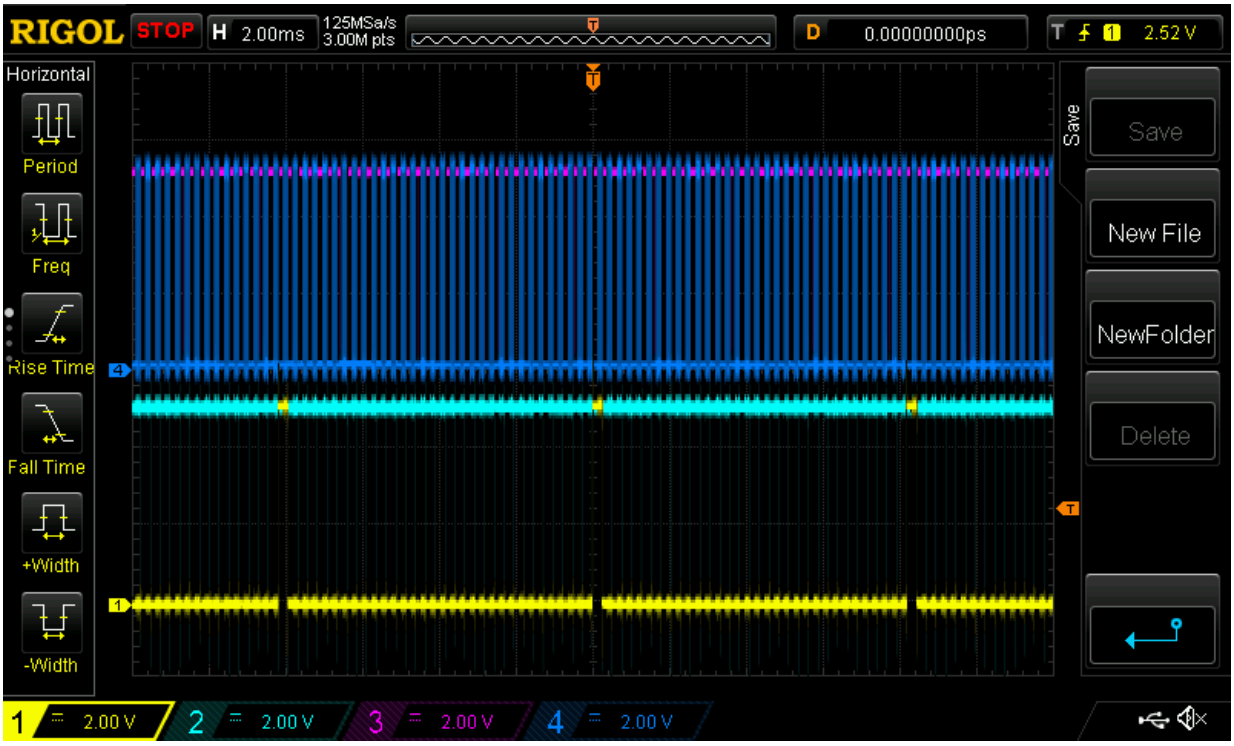


Figure 2. Rigol screenshot supplied with the pixel-data recording.

6. Successful tests and what each proves

CSV / test	Reported observation	Meaning
NewFile4.csv — attract-mode timing	128 DCLK rising edges between consecutive CL pulses in 46 complete intervals; ~255.98 microsecond row period.	Establishes serial width and row/latch timing without assuming pixel values.
NewFile5.csv — leftmost column	93 complete 128-clock groups; each decoded row had one illuminated bit in the first serial position.	Validates bit order and pixel polarity for a known vertical-line pattern.
NewFile6.csv — top horizontal row	96 identifiable groups; 3 groups had 128 lit pixels, 32 row intervals apart.	Validates top-row data alignment with RD marker.
NewFile7.csv — bottom horizontal row	93 complete groups; lit bursts at indices 14, 46 and 78, separated by 32 intervals.	Validates row progression and wraparound.

Illustrated reference images

Leftmost column



Top row



Bottom row



The images above are idealised illustrations of the test patterns, not photographs or plots directly rendered from the CSV samples. The matching patterns and counts were reported in the previous analysis.

7. Decoder firmware requirements — specification, not code

ID	Requirement	Acceptance criterion
CAP-01	Passive high-impedance input interface with safe 5 V to 3.3 V translation.	Original plasma DMD continues operating unchanged.
CAP-02	Hardware-assisted capture of DCLK edges and synchronous SD values.	128 bits per row at approximately 1 MHz without loss.
CAP-03	Detect RD frame boundary and align first following complete row as row 0.	Top/bottom diagnostic rows appear in correct positions.
CAP-04	Validate row count, DCLK count, and optional CL/RCLK timing.	Reject or flag incomplete/ambiguous rows and frames.
CAP-05	Publish immutable 512-byte packed frame with sequence number and timestamp.	No partially updated image displayed or hashed.
CAP-06	Render framebuffer to TTGO LCD using correct dimensions and scaling.	Three diagnostic patterns visibly match original DMD.
CAP-07	Calculate CRC32 over canonical row-major, MSB-first frame bytes.	Identical binary frames have identical CRC32.
CAP-08	Measure image duration and distinguish repeated refreshes from image changes.	Report image start, end and duration from timestamps.
CAP-09	Retain multiple raw frames for future brightness decoding.	Intermediate-intensity behaviour can be analysed offline.
CAP-10	Provide counters and raw-byte diagnostic output.	Lost clocks, sync errors, row counts and CRC changes visible.

Suggested processing pipeline

Electrical signals → input buffering → edge-aligned bit acquisition → row validation → RD-based frame alignment → 512-byte immutable frame → CRC32 + timestamp → LCD drawing + optional image catalogue.

The 512-byte buffer is small; reliable sampling and synchronisation are the real-time constraints. Interrupt handlers should only perform bounded ISR-safe operations. In particular, do not call blocking driver APIs such as i2s_read() from an interrupt handler. The exact ESP32 peripheral approach (I2S, SPI slave variant, RMT, parallel capture, or FPGA-assisted capture) remains to be validated against the measured signal timing.

8. Image identity, brightness and duration

Raw image ID: CRC32 over 512 canonical bytes. For example, if the leftmost-column image appears again after 10 seconds, its raw-frame CRC32 should match its earlier value. CRC32 is a fast fingerprint, not a mathematically unique identifier: different images can theoretically collide. Store the full 512-byte frame when collision-free comparison is needed.

Duration: when a new canonical image differs from the previous one, close the prior interval and record elapsed time using a monotonic timer. At ~122 Hz, raw frame boundaries are about 8.2 ms apart. Use consistent timestamps and account for dropped frames.

Brightness: the electrical interface supplies a binary pixel value during each refresh. Multi-level perceived brightness may arise from temporal modulation over several scans. A single 512-byte frame can therefore be insufficient to identify the stable perceived image. Preserve consecutive frames and investigate the modulation period before defining a brightness-aware image ID.

Example of temporal accumulation — illustrative only

Four consecutive refresh bits	Number of ON scans	Simple relative level
0000	0	0%
1000	1	25%
1010	2	50%
1111	4	100%

This four-refresh model is an explanatory example, not a verified Williams WPC intensity encoding. Actual phase, weighting, and modulation period remain open questions.

9. What went wrong in early ESP32 experiments

The TFT could display independent colour-test patterns, but live DMD image reconstruction remained blank. The reported “100% non-zero DMA bytes” was incorrectly treated as proof of valid pixels. It is not: repeated 0xFF, stale ring-buffer bytes or mis-sampled data can also produce 100%. Earlier software called i2s_read() inside interrupts, reconstructed rows from arbitrary ring positions, and reset a row counter using DE, a signal measured near the row rate rather than the RD frame rate. These are design risks, not reasons to repeat the electrical measurements.

10. Evidence classification and remaining gaps

Topic	Status	Next action
128 bits / row; 32 rows / frame	Strongly supported by reported oscilloscope analysis and matching patterns.	Preserve CSV and reproduce decoder statistics.
SD sampled on DCLK rising edge	Supported for static diagnostic patterns.	Check setup/hold margin across transitions and varied imagery.
RD frame alignment and row order	Supported by top/bottom line tests.	Verify precise pipeline offset using CL timing if needed.
CL polarity and exact relationship to RCLK	Partially measured in timing capture.	Extract edge timestamps with reproducible script.
Pin 13 function	Unresolved; comparable schematic labels NC.	Check connector continuity and source of measured activity.
Brightness / grayscale encoding	Not established.	Capture stable mid-brightness image over multiple frames.
ESP32 peripheral capture reliability	Not established.	Implement validated buffering and error diagnostics.
Stable image hashing	Specification defined; not implemented.	Define canonical bytes, brightness handling and collision policy.

11. Evidence register and reproducibility

Artifact	Source / purpose
NewFile4.csv	Four-channel timing capture: RD, RCLK, CL, DCLK; attract mode.
NewFile5.csv	Leftmost-column test: RD, RCLK, SD, DCLK.
NewFile6.csv	Top-horizontal-row test: RD, RCLK, SD, DCLK.
NewFile7.csv	Bottom-horizontal-row test: RD, RCLK, SD, DCLK.
Rigol screenshots	Visual record of acquisition settings and trace appearance.
Williams WPC / Babcock schematic image	Connector naming and electrical reference.

Reproducibility note: retain the original CSV files unchanged. Record their hashes, oscilloscope probe attenuation, acquisition mode, thresholds and decoder version alongside any later reconstructed images. The CSV format includes X/CH1–CH4 columns plus a header containing the time start and increment; the export may include malformed bytes/records, so parse defensively.

Reference materials: Williams WPC DMD schematics (user-supplied image); Vishay Dale APD-128G032 interface documentation (comparable interface, not proof of exact Babcock timing); project conversation measurements and uploaded Rigol recordings. Avoid treating third-party timing assumptions as measurements from this machine.

12. Recommended next milestone

Freeze the monochrome capture protocol, preserve the CSV recordings and a reproducible offline decoder, then build a minimal ESP32 diagnostic that shows validated rows, sync, frame count and CRC32. Once the three known patterns render correctly, study intermediate brightness and define a stable perceived-image identity. The electrical test sequence does not need repeating unless new evidence contradicts it.